

Error in the yield forecast of c-Si systems due to spectral effects

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Abstract — The current standard practice for forecasting the energy yield of a c-Si system includes the assumption that spectral effects are negligible. In this work, we compute the error that this assumption introduces. We conclude that the error in annual yield forecasts is up to $\pm 3\%$ but within $\pm 1.5\%$ for most locations. We also conclude that the variability due to daily and seasonal effects is $\pm 2\%$ to $\pm 3\%$ when the sun's zenith angle is less than 85° . With simulation, we demonstrate how a spectral correction factor f_λ can be determined for any system—including complicated installations such as bifacial single-axis trackers—and we describe how f_λ depends on latitude, zenith angle, albedo and several atmospheric conditions.

I. INTRODUCTION

The spectrum of the sunlight that falls on a PV module depends on the sun's position in the sky, the ground's albedo, and several atmospheric conditions [1, 2]. It changes throughout each day and year, and as a result, the efficiency of the modules in a system is usually either higher or lower than their nominal efficiency (the efficiency measured at standard test conditions). This change in efficiency is usually accounted for by multiplying the module's short-circuit current density J_{SC} under the AM1.5g spectrum by a spectral correction factor f_λ ,

$$J_{SC} = f_\lambda \times J_{SC \text{ AM1.5g}}. \quad (1)$$

When forecasting the yield of a system that contains c-Si modules, it is common practice to assume that $f_\lambda = 1$; i.e., to neglect spectral effects. For example, the default inputs in PVSyst (version 7.4.6) is $f_\lambda = 1$, and the help file states that spectral variation of c-Si modules is 'extremely weak' [3]. In fact, f_λ for annual yield tends to fall within the range 0.98–1.02, and daily and seasonal variations can also be significant [4].

One reason that f_λ is often neglected is that it has been difficult to determine f_λ at a particular site. Firstly, f_λ depends on atmospheric conditions like cloud cover, the concentrations of aerosols, water vapor and ozone, and the atmospheric pressure, all of which change from site to site and from day to day. Moreover, these atmospheric conditions affect direct and diffuse spectra differently, and hence one must compute the fractions of direct and diffuse light for a system's orientation (e.g., module tilt and azimuth), as well as the ground albedo in the case of bifacial systems.

Many works use either simulation or experiment to determine f_λ and parameterizations of f_λ [e.g., 4–9], and in so doing, they introduced some simplifications, such as neglecting one or more atmospheric effects, rear irradiance, temperature, etc.). Although such simplifications can be practical, they are not necessarily required.

In recent years there have been two major developments that assist researchers to attain a more accurate f_λ . Firstly, satellite measurements of atmospheric conditions are now readily accessible for any site in the world, and secondly, the advent of high-powered ray tracing has meant that yield simulations can be conducted with fewer simplifications.

In this work, we first describe the major factors that affect f_λ for a modern c-Si module, including their bounds and trends, and then we demonstrate how f_λ can be determined for any site—even complicated installations such as bifacial trackers with wavelength-dependent albedo. Finally, we estimate the error introduced by assuming that $f_\lambda = 1$. We intend to publish a longer and more detailed version of this conference paper in the near future.

II. GENERAL APPROACH

Spectral models require inputs for the air mass, the aerosol optical depth (AOD) at 500 nm, precipitable water vapor (PWV), ozone concentration, atmospheric pressure, and ground albedo. The values used to determine the direct, diffuse and global AM1.5 spectra [10] are listed in Table I.

TABLE I

PARAMETERS THAT AFFECT THE SOLAR SPECTRUM, THEIR APPROXIMATE RANGE OBSERVED OVER THE GLOBE, AND THE VALUES USED TO GENERATE THE AM1.5G SPECTRUM.

Parameter	Units	Range	AM1.5g
Cloud fraction		0–1	0
Air mass		1–30	1.5
AOD at 500 nm		0.05–3	0.084
PWV	cm	0.1–8	1.416
Ozone concentration	DU	200–470	343.8
Atmospheric pressure	mbar	650–1030	1013.25
Far-field albedo	%	5–90	Lt bare soil

We use those spectral models to compute the effect of atmospheric conditions on module behavior. This is performed in two ways:

Firstly, and as performed in many other studies [e.g., 4–9], the spectra are multiplied by the external quantum efficiency (EQE) of a typical c-Si module to determine the module's J_{SC} from front illumination. This method permits a rapid evaluation of how the atmospheric changes affect module power. The EQE used here is plotted in Fig. 1.

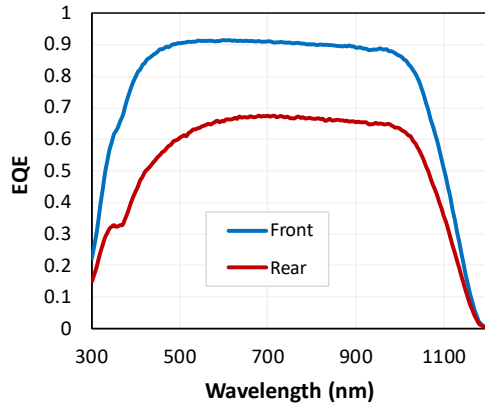


Fig. 1. Front and rear EQE of a modern c-Si module. This data used in the calculations of f_{λ} .

Secondly, the spectra are used to solve the hourly maximum module power P_{MP} and annual yield at a particular location, accounting for the position of the sun in the sky throughout the year, and many secondary effects, such as the impacts of temperature, electrical mismatch, bifacial irradiance, angular effects, structural shading, etc. These simulations are performed using SunSolve [11] with a single-axis tracker (SAT) configuration very similar to that published in [12].

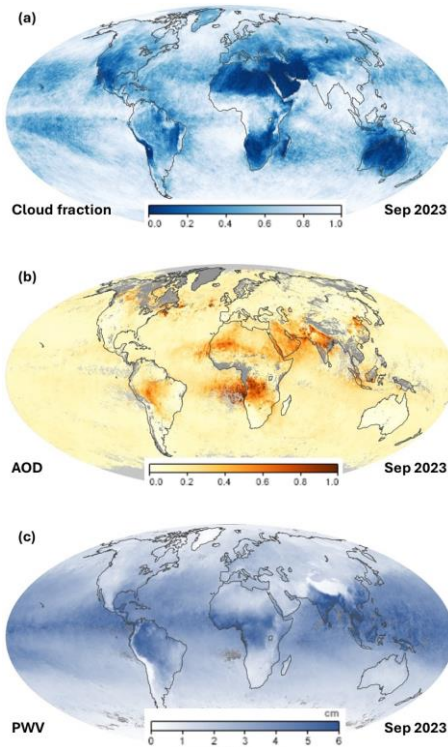


Fig. 2. Global maps of the average (a) cloud fraction, (b) AOD and (c) PWV in September 2023, as taken from [13] and [14].

III. THE EFFECT OF PWV ON SPECTRAL MISMATCH

Fig. 2 presents example global maps of the main atmospheric inputs: cloud fraction, AOD and PWV. To illustrate the role of atmosphere, Fig. 3 plots the clear-sky global spectral intensity to a horizontal plane for a range of PWV. The data is calculated for the default atmospheric conditions (Table I) using SPECTRL2 [2] over the range 280–4000 nm; for clarity, the spectra are plotted to 2500 nm.

Fig. 3 shows that an increase in PWV decreases the irradiance, primarily due to the deepening of H₂O absorption bands [1, 2]. It so happens that the reduction in irradiance at wavelengths below 1200 nm is less than the reduction in irradiance at wavelengths above 1200 nm. Consequently, as PWV increases, the power produced by a c-Si module decreases whereas its efficiency increases.

Fig. 4 quantifies this effect, plotting ΔJ_{SC} produced by the c-Si module for the case when the spectra are (a) unscaled and (b) scaled to an irradiance of 1000 W/m² (integrated over the range 280–4000 nm). Thus, under the same GHI, a module in a high PWV atmosphere will produce less current in the H₂O absorption bands but more current at other wavelengths. In short: f_{λ} increases as PWV increases.

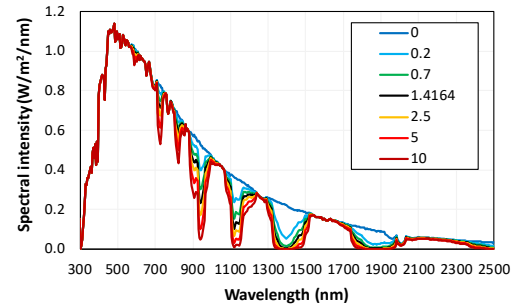


Fig. 3. The clear-sky global spectral intensity on a horizontal plane calculated for the default AM1.5 inputs and a range of PWV in cm.

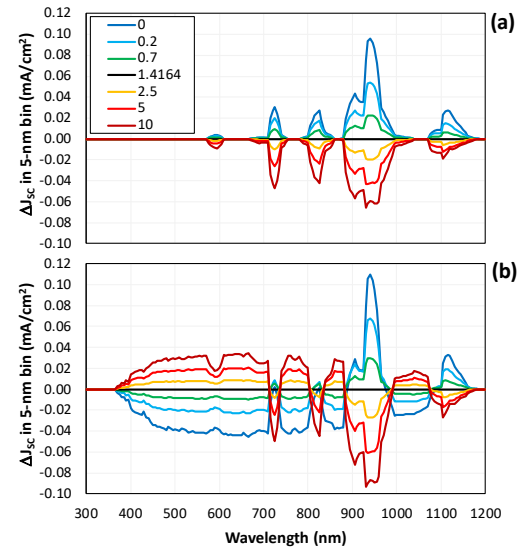


Fig. 4. Calculated ΔJ_{SC} of a horizontal c-Si module for the default AM1.5 inputs and a range of PWV (cm), where the clear-sky spectra are (a) unscaled, and (b) scaled to 1000 W/m².

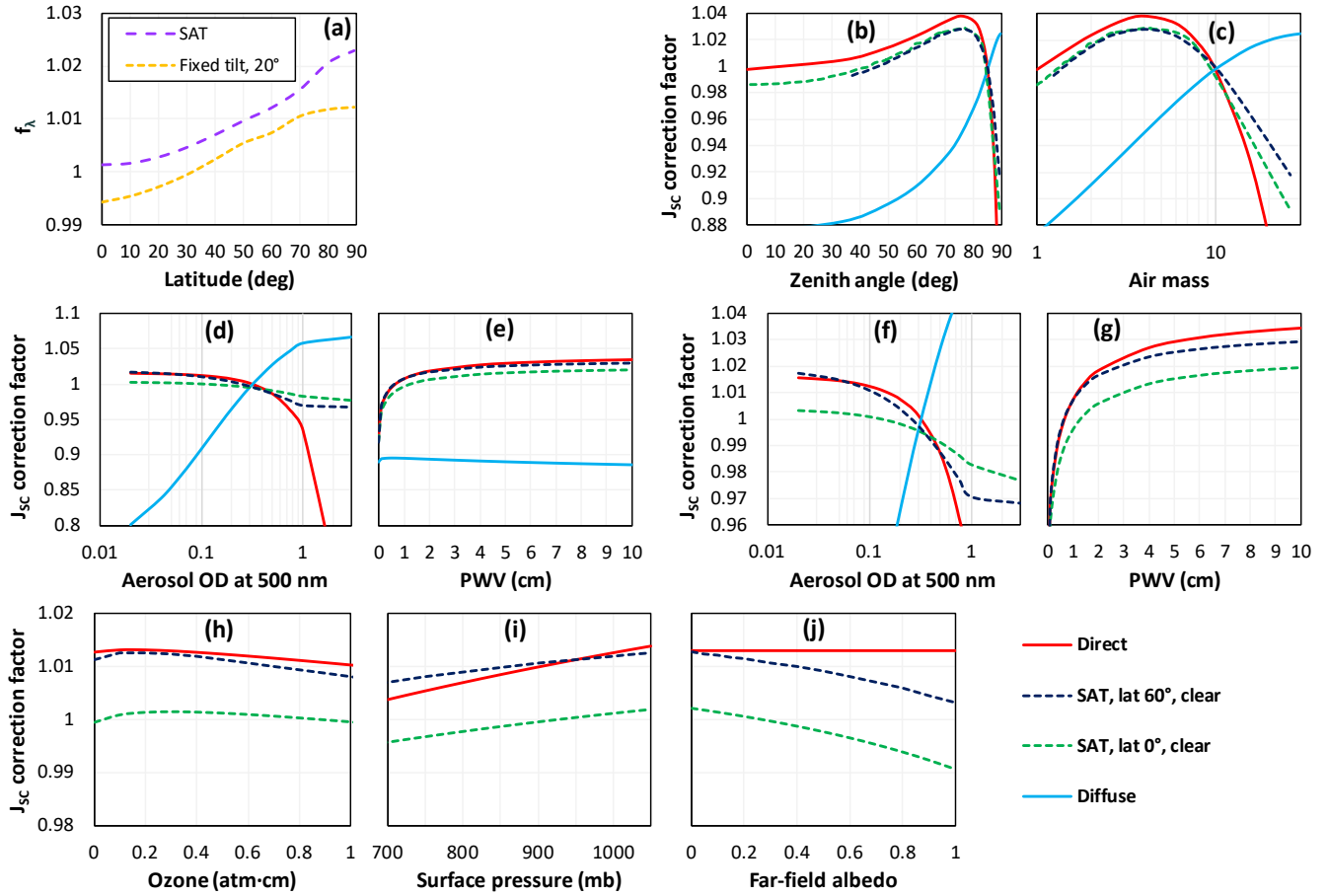


Fig. 5. f_{λ} vs (a) latitude, (b) zenith angle, (c) air mass, (d) and (f) AOD at 500 nm on different scales, (e) and (g) PWV, (h) ozone concentration, (i) surface pressure and (j) far-field albedo when all other inputs are held at the default values. f_{λ} plotted for perpendicular direct light (red), and isotropic diffuse light that is horizontal module (blue). Annual f_{λ} plotted for a particular latitude under clear skies.

IV. SPECTRAL CORRECTION FACTOR

Fig. 5 plots f_{λ} under clear-sky conditions when a single atmospheric input is varied from its default AM1.5g value. The red line plots f_{λ} for direct perpendicular light and the blue line plots f_{λ} for diffuse light incident to the horizontal. They exhibit opposing trends because atmospheric changes tend to convert direct photons into diffuse photons. They provide a guide to how f_{λ} depends on clear-sky atmospheric conditions for c-Si.

To examine the behavior of installed modules, the dashed lines in Fig 4 plot the annual f_{λ} of a monofacial SAT in clear-sky conditions at latitudes of 0° and 60°. These more practical f_{λ} are much closer to the red line than the blue line because there are many more photons in the direct clear-sky spectrum than in the diffuse clear-sky spectrum. We observe that f_{λ} increases as

- latitude increases,
- zenith angle increases (up to 75°),
- AOD decreases,
- PWV increases,
- ozone decreases (down to 200 DU),
- atmospheric pressure increases (elevation increases),
- far-field albedo decreases.

V. CASE STUDIES AND TYPICAL ERROR

The results of Fig. 5 illustrate important trends but omit many significant effects, including the impact of tracking, wavelength-dependent albedo, bifacial irradiance and clouds. These effects, and many secondary effects (like those relating to temperature, incident-angle, electrical mismatch, and non-linearities with irradiance), are all accounted for in a SunSolve-Yield simulation. At each time step, SunSolve first generates direct and diffuse spectra, then uses ray tracing to determine the spectra incident to the front and rear module surfaces, then computes the generation in each cell, and then solves the module J_{SC} via a SPICE model [12]. An annual yield in hourly intervals is solved in less than 2 minutes.

In these simulations, the clear-sky spectra were computed with the SPECTRL2 model [2] and the cloudy-sky spectra were determined by manipulating the clear-sky spectra using the model of Ernst *et al.* [15], which effectively assumes neutral density scattering and filtering by the clouds.

Fig. 6 plots the hourly, daily and annual f_{λ} for a bifacial SAT in Golden with the albedo of green rye grass. The resulting f_{λ} is 0.994 ± 0.021 (95% confidence interval). We found that at

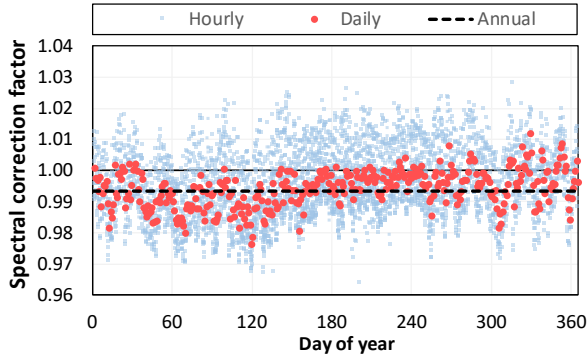


Fig. 6. f_{λ} for a bifacial SAT in Golden CO.

Golden, the seasonal effect due to changing airmass is dampened by the seasonal effect in PWV.

Fig. 7 plots f_{λ} of the same bifacial SAT at eight sites around the world. The sites were selected to provide a wide variety of cloud, AOD, PWV and latitude. The green symbols plot the annual f_{λ} , and the error bars represent the range of hourly f_{λ} following the approach taken for Fig. 6. The annual f_{λ} varies from 0.975 to 1.005, and the hourly variation in f_{λ} lies between ± 0.016 and ± 0.087 . The largest variation in f_{λ} occurs at sites with a high variation in cloud and AOD.

As mentioned above, these simulations computed the cloudy-sky spectra via the Ernst approach, which is easily implemented and quick to solve. A less accessible but significantly more accurate model is contained in FARMS-NIT [16], which accounts for the wavelength-dependent scattering and absorption of clouds. An initial comparison of the two approaches yielded an approximate correction to our data of $f_{\lambda} = f_{\lambda, \text{Ernst}} \times \{0.01 + 0.99 \cdot \cosh[0.7 \cdot (K_t - 0.84)]\}$, where K_t is the clearness index and equal to $I_H / [I_0 \cos(\theta_s)]$, θ_s is the solar zenith, and I_H and I_0 are the irradiance incident to the horizontal ground and upper atmosphere, respectively.

When this cloud-sky correction is applied, we attain the orange symbols in Fig. 7. This correction leads to a significant increase in f_{λ} at all sites, particularly at cloudy sites like Beijing, Agder and Singapore. Following the correction, we find that the f_{λ} varies from 0.98 to 1.03 at the eight sites.

IV. CONCLUSION

Given the vast array of locations, atmospheric conditions, configurations and modules for systems around the world, one cannot categorically determine f_{λ} for every possible site. But by evaluating the trends of f_{λ} with atmosphere (Fig. 5), the expected range of atmospheric conditions (Table I and Fig. 2), and the results of the detailed simulations (Fig. 7), we conclude that a modern c-Si system should have an annual f_{λ} of 1 ± 0.03 , where the uncertainty is nearer ± 0.015 at most sites. We also conclude that the variability due to daily and seasonal effects is about $\pm 2\%$ for clear sites but as high as $\pm 8\%$ for cloudy and turbid sites, and that the variability due to daily and seasonal effects is $\pm 2\%$ to $\pm 3\%$ when $\theta_s < 85^\circ$.

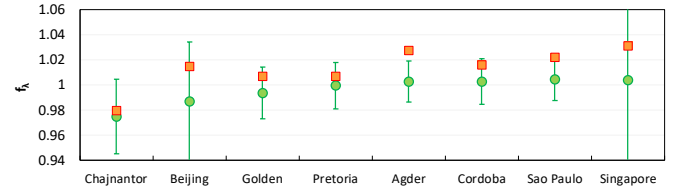


Fig. 7. f_{λ} at various sites. Green symbols plot annual f_{λ} , error bars plot the range of hourly f_{λ} (95% CI), and red symbols plot annual f_{λ} after applying a cloud correction devised from FARMS-NIT.

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